

COUNTING

Basic Counting Principles

THE PRODUCT RULE:

Suppose that a procedure can be broken down into a sequence of two tasks. If there are “ n_1 ” ways to do the first task and for each of these ways of doing the first task, there are “ n_2 ” ways to do the second task, then there are “ $n_1 n_2$ ” ways to do the procedure

EXAMPLE 1: A new company with just two employees rents a floor of a building with 12 offices. How many ways are there to assign different offices to these two employees?

Solution: The procedure of assigning offices to these two employees consists of assigning an office to first person, which can be done in 12 ways, then assigning an office to second person different from the office assigned to first person, which can be done in 11 ways. By the product rule, there are $12 \times 11 = 132$ ways to assign offices to these two employees.

EXAMPLE 2: The chairs of an auditorium are to be labeled with an uppercase English letter followed by a positive integer not exceeding 100. What is the largest number of chairs that can be labeled differently?

Solution: The procedure of labeling a chair consists of two tasks, namely, assigning to the seat one of the 26 uppercase English letters, and then assigning to it one of the 100 possible integers. The product rule shows that there are $26 \times 100 = 2600$ different ways that a chair can be labeled. Therefore, the largest number of chairs that can be labeled differently is 2600.

EXAMPLE 3: There are 32 computers in a data center in the cloud. Each of these computers has 24 ports. How many different computer ports are there in this data center?

Solution: The procedure of choosing a port consists of two tasks, first picking a computer and then picking a port on this computer. Because there are 32 ways to choose the computer and 24 ways to choose the port no matter which computer has been selected, the product rule shows that there are $32 \times 24 = 768$ ports.

Note: The product rule is often phrased in terms of sets in this way: If $A_1, A_2, \dots, A_m$ are finite sets, then the number of elements in the Cartesian product of these sets is the product of the number of elements in each set. To relate this to the product rule, note that the task of choosing an element in the Cartesian product $A_1 \times A_2 \times \dots \times A_m$ is done by choosing an element in A_1 , an element in $A_2, \dots$, and an element in A_m . By the product rule it follows that

$$|A_1 \times A_2 \times \dots \times A_m| = |A_1| \cdot |A_2| \cdots |A_m|.$$

EXAMPLE 4: How many different license plates can be made if each plate contains a sequence of three uppercase English letters followed by three digits (and no sequences of letters are prohibited, even if they are obscene)?

Solution: There are 26 choices for each of the three uppercase English letters and 10 choices for each of the three digits. Hence, by the product rule there are a total of $26 \cdot 26 \cdot 26 \cdot 10 \cdot 10 \cdot 10 = 17,576,000$ possible license plates.

THE SUM RULE:

If a task can be done either in one of n_1 ways or in one of n_2 ways, where one of the set of n_1 ways is the same as any of the set of n_2 ways, then there are $n_1 + n_2$ ways to do the task.

EXAMPLE 1: Suppose that either a member of the mathematics faculty or a student who is mathematics major is chosen as a representative to a university committee. How many different choices are there for this representative if there are 37 members of the mathematics faculty and 83 mathematics majors and no one is both a faculty member and a student?

Solution: There are 37 ways to choose a member of the mathematics faculty and there are 83 ways to choose a student who is mathematics major. Choosing a member of the mathematics faculty is never the same as choosing a student who is mathematics major because no one is both a faculty member and a student. By the sum rule it follows that there are $37 + 83 = 120$ possible ways to pick this representative.

Note: Suppose that a task can be done in one of n_1 ways, in one of n_2 ways, $\dots$, or in one of n_m ways, where none of the set of n_i ways of doing the task is the same as any of the set of n_j ways, for all pairs i and j with $1 \leq i < j \leq m$. Then the number of ways to do the task is $n_1 + n_2 + \dots + n_m$.

EXAMPLE 2: A student can choose a computer project from one of three lists. The three lists contain 23, 15, and 19 possible projects, respectively. No project is on more than one list. How many possible projects are there to choose from?

Solution: The student can choose a project by selecting a project from the first list, the second list, or the third list. Because no project is on more than one list, by the sum rule there are $23 + 15 + 19 = 57$ ways to choose a project.

THE SUBTRACTION RULE:

If a task can be done in either n_1 ways or n_2 ways, then the number of ways to do the task is $n_1 + n_2$ minus the number of ways to do the task that are common to the two different ways.

Note: The subtraction rule is also known as the **principle of inclusion–exclusion**, especially when it is used to count the number of elements in the union of two sets. Suppose that A_1 and A_2 are sets. Then, there are $|A_1|$ ways to select an element from A_1 and $|A_2|$ ways to select an element from A_2 . The number of ways to select an element from A_1 or from A_2 , that is, the number of ways to select an element from their union, is the sum of the number of ways to select an element from A_1 and the number of ways to select an element from A_2 , minus the number of ways to select an element that is in both A_1 and A_2 . Because there are $|A_1 \cup A_2|$ ways to select an element in either A_1 or in A_2 , and $|A_1 \cap A_2|$ ways to select an element common to both sets, we have $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$.

EXAMPLE 1: A computer company receives 350 applications from college graduates for a job planning a line of new web servers. Suppose that 220 of these applicants majored in computer science, 147 majored in business, and 51 majored both in computer science and in business. How many of these applicants majored neither in computer science nor in business?

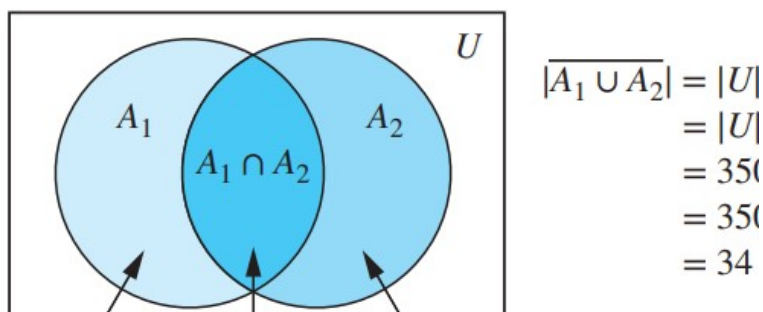
Solution: To find the number of these applicants who majored neither in computer science nor in business, we can subtract the number of students who majored either in computer science or in business (or both) from the total number of applicants.

Let A_1 be the set of students who majored in computer science i.e. 220 and A_2 the set of students who majored in business i.e. 147. Then $A_1 \cup A_2$ is the set of students who majored in computer science or business (or both), and $A_1 \cap A_2$ is the set of students who majored both in computer science and in business i.e. 51.

By the subtraction rule the number of students who majored either in computer science or in business (or both) equals $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2| = 220 + 147 - 51 = 316$.

We conclude that $350 - 316 = 34$ of the applicants majored neither in computer science nor in business.

A Venn diagram for this example is shown below



EXAMPLE 2: In a discrete mathematics class every student is a major in computer science or mathematics, or both. The number of students having computer science as a major (possibly along with mathematics) is 25; the number of students having mathematics as a major (possibly along with computer science) is 13; and the number of students majoring in both computer science and mathematics is 8. How many students are in this class?

Solution: Let A be the set of students in the class majoring in computer science and B be the set of students in the class majoring in mathematics.

Then $A \cap B$ is the set of students in the class who are joint mathematics and computer science majors. Because every student in the class is majoring in either computer science or mathematics (or both), it follows that the number of students in the class is $|A \cup B|$.

Therefore,

$$|A \cup B| = |A| + |B| - |A \cap B| = 25 + 13 - 8 = 30.$$

Therefore, there are 30 students in the class.

EXAMPLE 3: Among 50 patients admitted to a hospital, 25 are diagnosed with pneumonia, 30 with bronchitis, and 10 with both pneumonia and bronchitis.

Determine: (a) The number of patients diagnosed with pneumonia or bronchitis (or both).

(b) The number of patients not diagnosed with pneumonia or bronchitis.

Solution:

Let U denote the entire set of patients. Let P and B denote the set of patients diagnosed with pneumonia and bronchitis respectively. Thus: $|U| = 50$, $|P| = 25$, $|B| = 30$, $|P \cap B| = 10$.

(a) By using the inclusion-exclusion principle: $|P \cup B| = |P| + |B| - |P \cap B| = 25 + 30 - 10 = 45$. Thus 45 patients are diagnosed with pneumonia or bronchitis.

(b) This is the same as asking to determine $|\overline{P \cup B}|$. We know that there are 50 patients altogether – of which 45 are diagnosed with pneumonia or bronchitis. Use this to solve the question: $|U| = 50$, $|P \cup B| = 45$

Therefore,

$$|\overline{P \cup B}| = 50 - 45 = 55 \text{ patients are not diagnosed with pneumonia or bronchitis.}$$

THE DIVISION RULE:

There are n/d ways to do a task if it can be done using a procedure that can be carried out in n ways, and for every way w , exactly d of the n ways correspond to way w .

EXAMPLE 1: Suppose that an automated system has been developed that counts the legs of cows in a grass land. Suppose that this system has determined that in a farmer's grass land there are exactly 572 legs. How many cows are there in this grass land, assuming that each cow has four legs and that there are no other animals present?

Solution: Let n be the number of cow legs counted in a pasture. Because each cow has four legs, by the division rule we know that the pasture contains $n/4$ cows. Consequently, the pasture with 572 cow legs has $572/4 = 143$ cows in it.

EXAMPLE 2: How many different ways are there to seat four people around a circular table, where two seating's are considered the same when each person has the same left neighbor and the same right neighbor?

Solution: We arbitrarily select a seat at the table and label it seat 1. We number the rest of the seats in numerical order, proceeding clockwise around the table. Note that there are four ways to select the person for seat 1, three ways to select the person for seat 2, two ways to select the person for seat 3, and one way to select the person for seat 4. Thus, there are $4! = 24$ ways to order the given four people for these seats. However, each of the four choices for seat 1 leads to the same arrangement, as we distinguish two arrangements only when one of the people has a different immediate left or immediate right neighbor. Because there are four ways to choose the person for seat 1, by the division rule there are $24/4 = 6$ different seating arrangements of four people around the circular table.

Permutation and Combination

Permutation is an arrangement of objects in which the order does matter. The number of permutation of “n” objects chosen “r” objects at a time is denoted by ${}^n P_r$, where $0 < r \leq n$. A permutation of n objects taken r at a time is also called r-arrangement. This also represented as $P(n, r)$

- The number of permutations of “n” objects taken “r” at a time is given by ${}^n P_r = \frac{n!}{(n-r)!}$
- The number of permutations of “n” objects taken “r” at a time where repetition allowed is given by n^r .
- The number of permutations of “n” objects, where p objects are of the same kind and all different $= \frac{n!}{p!}$
- The number of permutations of “n” objects, where p_1 objects are of the first kind, p_2 are of the second kind, p_k of the k^{th} kind and the rest if any, are all of different kind is =

$$\frac{n!}{p_1! p_2! \dots p_k!}$$
- The of permutations of “n” objects taken “r” at a time is given by $n(n-1)(n-2) \dots (n-r+1)$ which is denoted by ${}^n P_r$ (when the repition is not allowed)
- ${}^n P_n = n!$
- ${}^n P_0 = 1$
- ${}^n P_1 = n$

Problem 1: How many four digit numbers can be formed by using the digits 1 to 9

- a) If repetition of the digit is allowed.
- b) If repetition of the digit is not allowed

Solution:

a) There are 9 numbers 1 to 9 and we have to form 4 digit number $n = 9$ and $r = 4$

When reptetion is allowed the number of permutations is $n^r = 9^4 = 6561$.

b) There are 9 numbers and 4 digits number should be formed

$n=9$ and $r=4$

Here repetition is not allowed

$$\therefore {}^n P_r = {}^9 P_4 = \frac{9!}{(9-4)!} = \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{5 \times 4 \times 3 \times 2 \times 1}$$

$$= 9 \times 8 \times 7 \times 6$$

$$= 3,024$$

Problem 2: Find the number of permutations of English vowels A, E, I, O, U taking two at a time. Also verify your result.

Solution: Number of permutations of A, E, I, O, U taking 2 at a time $= {}^5 P_2 = \frac{5!}{(5-2)!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1}$

$$= 5 \times 4 = 20$$

Verification: The required permutations are AE, AI, AO, AU, EA, EI, EO, EU, IA, IE, ID, IU, OA, OE, OI, OU, UA, UE, UI, UO. These are 20 in numbers.

Problem 3: Six candidates are called for an interview to fill up four posts in an office. Assuming that each candidate is fit for each post, determine the number of ways in which:

- (1) First and second posts can be filled up
- (ii) First three posts can be filled up
- (ii) All the four posts can be filled up

Solution: Number of candidates = 6

(1) Number of ways of filling first and second posts

= Number of permutations of 6 candidates taking 2 at a time

$$= {}^6P_2 = \frac{6!}{(6-2)!} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1} = 6 \times 5 = 30$$

(ii) Number of ways of filling first three posts

= Number of permutations of 6 candidates taking 3 at a time

$$= {}^6P_3 = \frac{6!}{(6-3)!} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1} = 6 \times 5 \times 4 = 120$$

(iii) Number of ways of filling all the four posts

= Number of permutations of 6 candidates taking 4 at a time

$$= {}^6P_4 = \frac{6!}{(6-4)!} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1} = 6 \times 5 \times 4 \times 3 = 360.$$

Problem 4: Find the number of ways in which 8 boys and 5 girls can be arranged in a row so that no two girls are together.

Solution:

Since no two girls can sit together, we first arrange the boys

8 boys can be arranged in ${}^8P_8 = 8! = 40320$ ways

X B X B X B X B X B X B X B X

When boys occupy seats corresponding to each sitting of boys, there are 9 places marked X for

girls. 5 girls can be arranged in these 9 places in ${}^9P_5 = \frac{9!}{(9-5)!} = \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1}$

$$= 9 \times 8 \times 7 \times 6 \times 5 = 15120 \text{ ways.}$$

$\therefore$ Corresponding to each of 40320 ways of arranging boys, the girls can be arranged in 15120 ways.

Total Number of arrangements = $40320 \times 15120 = 60,96,38,400$.

Problem 5: Three married couples are to be seated in a row having 6 seats in a cinema hall. If spouses are to be seated next to each other, in how many ways can they be seated? Find also the number of ways of their seating if all the ladies sit together.

Solution:

Number of couples = 3

1) Number of ways of arranging 3 couples = $3! = 6$

Number of ways of arranging spouses in each couple = $2! = 2$

Total No. of arrangements = $6 \times 2 \times 2 \times 2 = 48$

2) Since all ladies must sit together, we tie the ladies together and consider the group of ladies as one object.

Number of ways of arranging the group of ladies and 3 gets = $(1 + 3)! = 4! = 24$

In each of the above arrangements, ladies can be arranged themselves in $3!$, i.e. 6 ways.

Total number of arrangements = $24 \times 6 = 144$.

Problem 6: In how many ways words with or without meaning can be formed with the letters of the word EQUATION, vowels and consonants occur together?

Solution:

The word EQUATION, consists of 8 different letters, out of which E, U, A, I, O are vowels and Q, T, N are consonants.

In the required word, we tie the vowels together and the consonants together. The two group can be arranged in $2!$ ways. Corresponding to each of these arrangements, the 5 vowels can be arranged in $5!$ ways

Total number of words = $2 \times 5! \times 3! = 1440$.

Problem 7: How many words with or without meaning can be made from the letters of the word MONDAY assuming that no letter is repeated, if:

(i) 4 letters are used at a time

(ii) all letters are used at a time

(iii) all letters are used but first letter is a vowel?

Solution

The total number of letters in the word MONDAY is 6 and all are different.

(1) Words of 4 letters

Number of words of 4 letters = ${}^6P_4 = 6 \times 5 \times 4 \times 3 = 360$

(2) Words of all letters

Number of words of using all letters = ${}^6P_6 = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$

(iii) Words beginning with vowels. In the letters M, O, N, D, A, Y only O and A are vowels. In the required words, the first letter can be filled in ${}^2P_1 = 2$ ways (any one of O, A) and the remaining 5 letters in ${}^5P_5 = 5! = 120$ ways (4 consonants vowels)

By fundamental principle of counting, total number of words = $2 \times 120 = 240$.

Problem 8: Find how many arrangements can be made with the letters of the word MATHEMATICS. In how many of them the vowels occur together?

Solution

There are 11 letters (not all different) in the word MATHEMATICS

H occurs twice, A occurs twice. T occurs twice and the rest are all different.

$$\text{Total number of arrangements} = \frac{(11!)}{(2!)(2!)(2!)} = 4989600.$$

The vowels in the word are A, E, A, I. Treat these four vowels as one object. Then, we have to find the arrangements of 8 objects (7 consonants and one object AEAI). The consonants M and T occur twice and all other $8 - 4 = 4$ are different.

$$\text{Number of arrangements of 8 objects} = \frac{(8!)}{(2!)(2!)} = 10080.$$

The four vowels in the single object (AEAI) can also be arranged in $= \frac{(4!)}{(2!)} = 12$ ways.

$$\text{Required number of arrangements} = 10080 \times 12 = 120960.$$

Problem 9: How many different words can be formed with the letters of the word HARYANA?

How many of these:

- (i) Have H and N together?
- (ii) Have three vowels together?
- (ii) Begin with H and end with N?

Solution:

(i) In the word HARYANA, there are 7 letters. A occurs thrice and the rest are all different.

$$\text{Total number of words} = \frac{7!}{3!} = \frac{5040}{6} = 840$$

In this case, we consider the pair (HN) as one object.

Number of objects to be arranged is 6 in which A is repeated thrice.

$$\text{Number of arrangements} = \frac{6!}{3!} = \frac{720}{6} = 120$$

H and N can also be arranged among themselves in $2! = 2$ ways

$$\text{Total number of words in which H and N are together} = 120 \times 2 = 240$$

(ii) In words beginning with H and ending with N, we have to fill 5 places with 5 letters A, A, A, R, Y

$$\text{Total number of words} = 1 \times 1 \times \frac{5!}{3!} = \frac{120}{6} = 20$$

(iii) The three vowels in the word HARYANA are A, A, A

We have to arrange 5 objects (AAA), H, R, Y, N and this can be done in $5! = 120$ ways
Total numbers of words = 120.

Problem 10: In how many ways can the letters of the word ASSASSINATION be arranged so that all the S's are together.

Solution:

In the given word ASSASSINATION, the letter A appears 3 times, S appears 4 times, I appears 2 times, N appears 2 times and all the other letters appear only once. Since all the words have to be arranged in such a way that all the S's are together, SSSS is treated as a single object for the time being. This single object together with the remaining objects will account for 10 objects.

$$\text{These 10 objects in which there are 3 A's, 2 I's and 2 N's ways can be arranged in } \frac{10!}{3!2!2!}$$

ways

A combination is a group of outcomes in which the order does not matter.

The number of combinations of “n” objects taken “r” at a time is denoted by nC_r , where $0 < r \leq n$.

The number of all combinations of n objects taken r at a time is denoted by nC_r or $C(n, r)$ or $\binom{n}{r}$

- The number of combinations of “n” objects taken “r” at a time is denoted by ${}^nC_r = \frac{n!}{(n-r)!r!}$
- ${}^nC_n = 1$
- ${}^nC_0 = 1$
- ${}^nC_1 = n$
- ${}^nC_r = {}^nC_{n-r}$

Problem 1: A committee of 7 has to be formed from 9 boys and 4 girls. In how many ways can this be done when the committee consists of:

- exactly 3 girls
- at least 3 girls
- at most 3 girls.

Solution

A committee of 7 has to be formed from 9 boys and 4 girls.

- Since exactly 3 girls are to be there in every committee, each committee must consists of $(7-3) = 4$ boys only.

Thus, in this case, required number of ways are $= {}^4C_3 \times {}^9C_4$

$$\begin{aligned}
 &= \frac{4!}{(4-3)!3!} \times \frac{9!}{(9-4)!4!} \\
 &= \frac{4!}{1!3!} \times \frac{9!}{5!4!} \\
 &= \frac{4 \times 3 \times 2 \times 1}{3 \times 2 \times 1} \times \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1 \times 5 \times 4 \times 3 \times 2 \times 1} \\
 &= 504
 \end{aligned}$$

- Since at least 3 girls are to be there in every committee, the committee can consist of

(a) 3 girls and 4 boys or (b) 4 girls and 3 boys

3 girls and 4 boys can be selected in ${}^4C_3 \times {}^9C_4$ ways

$$\begin{aligned}
 &= \frac{4!}{(4-3)!3!} \times \frac{9!}{(9-4)!4!} \\
 &= \frac{4!}{1!3!} \times \frac{9!}{5!4!} \\
 &= \frac{4 \times 3 \times 2 \times 1}{3 \times 2 \times 1} \times \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{5 \times 4 \times 3 \times 2 \times 1 \times 4 \times 3 \times 2 \times 1} \\
 &= 504
 \end{aligned}$$

4 girls and 3 boys can be selected in ${}^4C_4 \times {}^9C_3$ ways

$$\begin{aligned}
 &= \frac{4!}{(4-4)!4!} \times \frac{9!}{(9-3)!3!} \\
 &= \frac{4!}{0!4!} \times \frac{9!}{6!3!} \\
 &= \frac{4 \times 3 \times 2 \times 1}{1 \times 4 \times 3 \times 2 \times 1} \times \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{6 \times 5 \times 4 \times 3 \times 2 \times 1 \times 3 \times 2 \times 1} = 84
 \end{aligned}$$

Therefore, in this case, required number of ways are ${}^4C_3 \times {}^9C_4 + {}^4C_4 \times {}^9C_3 = 504 + 84 = 588$

(iii) Since at most 3 girls are to be there in every committee, the committee can consist of

(a) 3 girls and 4 boys

(b) 2 girls and 5 boys

(c) 1 girls and 6 boys

(d) No girl and 7 boys

3 girls and 4 boys can be selected in ${}^4C_3 \times {}^9C_4$ ways

$$\begin{aligned} &= \frac{4!}{(4-3)!3!} \times \frac{9!}{(9-4)!4!} \\ &= \frac{4!}{1!3!} \times \frac{9!}{5!4!} \\ &= \frac{4 \times 3 \times 2 \times 1}{3 \times 2 \times 1} \times \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1 \times 5 \times 4 \times 3 \times 2 \times 1} \\ &= 504 \end{aligned}$$

2 girls and 5 boys can be selected in ${}^4C_2 \times {}^9C_5$ ways

$$\begin{aligned} &= \frac{4!}{(4-2)!2!} \times \frac{9!}{(9-5)!5!} \\ &= \frac{4!}{2!2!} \times \frac{9!}{4!5!} \\ &= \frac{4 \times 3 \times 2 \times 1}{2 \times 1 \times 2 \times 1} \times \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1 \times 5 \times 4 \times 3 \times 2 \times 1} \\ &= 756 \end{aligned}$$

1 girls and 6 boys can be selected in ${}^4C_1 \times {}^9C_6$ ways

$$\begin{aligned} &= \frac{4!}{(4-1)!1!} \times \frac{9!}{(9-6)!6!} \\ &= \frac{4!}{3!1!} \times \frac{9!}{3!6!} \\ &= \frac{4 \times 3 \times 2 \times 1}{3 \times 2 \times 1 \times 1} \times \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1} \\ &= 336 \end{aligned}$$

No girls and 7 boys can be selected in ${}^4C_0 \times {}^9C_7$ ways.

$$\begin{aligned} &= \frac{4!}{(4-0)!0!} \times \frac{9!}{(9-7)!7!} \\ &= \frac{4!}{4!0!} \times \frac{9!}{2!7!} \\ &= \frac{4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1 \times 1} \times \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1} \\ &= 36 \end{aligned}$$

Therefore, in this case, required number of ways are ${}^4C_3 \times {}^9C_4 + {}^4C_2 \times {}^9C_5 + {}^4C_1 \times {}^9C_6 + {}^4C_0 \times {}^9C_7 = 504 + 756 + 336 + 36 = 1632$.

Problem 2: It is compulsory to answer 10 questions in an examination choosing at least 4 questions from each part A and part B. If there are 6 questions in part A and 7 questions in part B, in how many ways can 10 questions be attempted?

Solution:

Case I: 4 questions from part A and 6 questions from part B

Number of ways of choosing 4 questions from part A = ${}^6C_4 = 15$

Number of ways of choosing 6 questions from part B = ${}^7C_6 = 7$

Total number of ways = $15 \times 7 = 105$

Case II: 5 questions from part A and 5 questions from part B

Number of ways of choosing 5 questions from part A = ${}^6C_5 = 6$

Number of ways of choosing 5 questions from part B = ${}^7C_5 = 21$

Total number of ways = $6 \times 21 = 126$

Case II: 6 questions from part A and 4 questions from part B

Number of ways of choosing 6 questions from part A = ${}^6C_6 = 1$

Number of ways of choosing 4 questions from part B = ${}^7C_4 = 35$

Total number of ways = $1 \times 35 = 35$

Required number of ways = $105 + 126 + 35 = 266$.

Problem 3: How many ways a 5-member committee can be formed out of 10 people if two particular people must be included?

Solution:

Number of people to be selected if two particular people must be included = $5 - 2 = 3$

Number of ways of selecting 3 out of $(10 - 2) = 8$ people = ${}^8C_3 = \frac{8!}{3! \times 5!} = 56$.

$\therefore$ there are 56 ways of forming such a committee.

Problem 4: There are 4 balls of color red, green, yellow and blue. In how many ways 2 two balls can be selected such that one of them is red or blue?

Solution:

Number of ways two balls can be selected out of 4 balls = ${}^4C_2 = \frac{4!}{2! \times 2!} = 6$ ways

Number of ways two balls can be selected such that neither red nor blue ball gets selected =

$$({}^{4-2})C_2 = {}^2C_2 = 1$$

Number of ways two balls can be selected such that either a red or a blue ball gets selected = $6 - 1 = 5$ ways.

Problem 5: A team of four has to be selected from 6 boys and 4 girls. How many different ways a team can be selected if at least one boy must be there in the team?

Solution:

Combination of a four-member team with at least one boy are:

$\{(BGGG), (BBGG), (BBBG), (BBBB)\}$

Number of ways one boy and three girls can be selected = ${}^6C_1 \times {}^4C_3 = 6 \times 4 = 24$

Number of ways two boys and two girls can be selected $= {}^6C_2 \times {}^4C_2 = 15 \times 6 = 90$

Number of ways three boys and one girl can be selected $= {}^6C_3 \times {}^4C_1 = 20 \times 4 = 80$

Number of ways four boys can be selected $= {}^6C_4 = 15$

Total number of ways to form such a team $= 24 + 90 + 80 + 15 = 209$.

Problem 6: How many ways are there to select five players from a 10-member tennis team to make a trip to a match at another school?

Solution: The answer is given by the number of 5-combinations of a set with 10 elements.

The number of such combinations is $= {}^{10}C_5 = \frac{10!}{5! 5!} = 252$.

Problem 7: Suppose that there are 9 faculty members in the mathematics department and 11 in the computer science department. How many ways are there to select a committee to develop a discrete mathematics course at a school if the committee is to consist of three faculty members from the mathematics department and four from the computer science department?

Solution: By the product rule, the answer is the product of the number of 3-combinations of a set with nine elements and the number of 4-combinations of a set with 11 elements.

The number of ways to select the committee is ${}^9C_3 \times {}^{11}C_4 = \frac{9!}{6! 3!} \times \frac{11!}{7! 4!} = 84 \times 330 = 27,720$

Pigeonhole principle and Generalized pigeonhole principle

Pigeonhole principle:

If m objects (pigeons) are placed into n containers (pigeonholes), then at least one container (pigeonholes) must contain more than one object (pigeons) in it.

Proof: By contradiction method

Consider m objects (pigeons) and n containers (pigeonholes), where $m > n$.

Assumption: Suppose that no container(pigeonholes) holds more than one object. This would mean that each container (pigeonholes), contains at most one object(pigeons).

Consequences: If each container(pigeonholes), holds at most one object(pigeons), then the maximum number of objects(pigeons) that can be placed in the n containers(pigeonholes), is n . That is, the total number of objects (pigeons) cannot exceed n , because each container(pigeonholes), can hold only one object(pigeons).i.e. $m \leq n$.

This contradicts the fact that there are m objects(pigeons), and we are assuming that $m > n$.

Conclusion: Therefore, our assumption that no container(pigeonholes), holds more than one object must be false. Hence, there must be at least one container(pigeonholes), that holds more than one object.

Generalized pigeonhole principle:

If m objects (pigeons) are placed into n containers (pigeonholes), then at least one container (pigeonholes) must contain $(p+1)$ or more object (pigeons) in it,

where $p = \left\lfloor \frac{m-1}{n} \right\rfloor$.

Proof: By contradiction method

Consider m objects (pigeons) and n containers (pigeonholes).

Assumption: Suppose that no container(pigeonholes) holds $(p+1)$ or more object. This would mean that each container (pigeonholes), contains p or less number of object(pigeons).

Consequences: If each container(pigeonholes), holds at most p object, then the maximum number of objects that can be placed in the n containers(pigeonholes), is p or less number of object(pigeons).

Then,

The total number of objects (pigeons) $\leq np = n \left\lfloor \frac{m-1}{n} \right\rfloor$

$$\leq n \left(\frac{m-1}{n} \right) = m-1$$

This means the total number of objects (pigeons) = $m-1$

This contradicts the fact that there are m objects (pigeons).

Conclusion: Therefore, our assumption that no container(pigeonholes), holds $(p+1)$ or more object. Hence, there must be at least one container(pigeonholes), that holds $(p+1)$ or more object(pigeons).

Example 1: Among any group of 367 people there must be at least two with the same birthday because there are only 365 possible birthdays.

Example 2: In any group of 27 English words, there must be at least two that begin with the same letter, because there are 26 letters in the English alphabet.

Example 3:

How many students must be in a class to, guarantee that at least two students receive the same score on the final exam, if the exam is graded on a scale from 0 to 100 points?

Solution:

There are 101 possible scores on the final. The pigeonhole principle shows that among any 102 students there must be at least 2 students with the same score.

Example 4: Suppose a department contains 13 professors. Then two of the professors were born in the same month.

There are 12 months in a year, and 13 professors are there. According to pigeon hole principle if we assume each professor born in different month then 12 professor born in 12 months. So, according to pigeonhole principle, two of the professors were born in the same month.

Example 5: Suppose a laundry basket contains many red, white and blue socks. Then one need only grab four socks to be sure of getting a pair with same color.

We have to grab 4 socks. Two socks should be of color and we have 3 different colors.

If we choose 4 socks, definitely there will be two socks of same color.

Example 6: A drawer contains ten black and ten white socks. What is the least number of socks we must pull out to be sure to get a matched pair?

If we pick just two socks, they may have different colors. But when we pick a third sock, it must be the same color as one of the socks already chosen. Hence the answer is three.

Example 7:

Assume 100 people .can you tell something about the number of people born in the same month?

Solution:

Given:

- $m = 100$
- $n = 12$

We are looking for the minimum number of people that must be in born in the same month

By **generalized pigeonhole principle**

$$\begin{aligned} p+1 &= \left\lfloor \frac{m-1}{n} \right\rfloor + 1 \\ &= \left\lfloor \frac{100-1}{12} \right\rfloor + 1 \\ &= \left\lfloor \frac{99}{12} \right\rfloor + 1 = [8.25] + 1 = 8+1=9 \end{aligned}$$

Among 100 people there are at least 9 who were born in the same month.

Example 8:

Show that among any set of 5 integers, there are 2 with the same remainder when divided by 4.

Solution:

Given:

- $m = 5$
- $n = 4$

By **generalized pigeonhole principle**

$$\begin{aligned} p+1 &= \left\lfloor \frac{m-1}{n} \right\rfloor + 1 \\ &= \left\lfloor \frac{5-1}{4} \right\rfloor + 1 \\ &= \left\lfloor \frac{4}{4} \right\rfloor + 1 = \lfloor 1 \rfloor + 1 = 1 + 1 = 2 \end{aligned}$$

Example 9:

Suppose we have 25 items and 6 containers. What is the minimum number of items that must be in at least one container?

Solution:

Given:

- $m=25$
- $n=6$

We are looking for the minimum number of items that must be in at least one container.

The **generalized pigeonhole principle** tells us that if the items are distributed as evenly as possible, at least one container will have at least:

$$\begin{aligned} p+1 &= \left\lfloor \frac{m-1}{n} \right\rfloor + 1 \\ &= \left\lfloor \frac{25-1}{6} \right\rfloor + 1 \\ &= \left\lfloor \frac{24}{6} \right\rfloor + 1 = \lfloor 4 \rfloor + 1 = 4 + 1 = 5. \end{aligned}$$

Thus, at least one container will have at least 5 items.

Example 10:

There are 50 students, and we want to divide them into 8 groups. What is the smallest number of students that must be in at least one group?

Solution:

Given:

- $m=50$
- $n=8$.

The **generalized pigeonhole principle** tells us that the smallest number of students that must be in at least one group is:

$$\begin{aligned} p+1 &= \left\lfloor \frac{m-1}{n} \right\rfloor + 1 \\ &= \left\lfloor \frac{50-1}{8} \right\rfloor + 1 \\ &= \left\lfloor \frac{49}{8} \right\rfloor + 1 = \lfloor 6.125 \rfloor + 1 = 6 + 1 = 7. \end{aligned}$$

Thus, at least one group must contain at least 7 students.